DGMS JSO GEOLOGY 2019 • APPLIED GEOPHYSICS & MINERAL EXPLORATION

Question 99: Geophysical Methods: Units, Density Contrasts & Magnetics

Systematic Analysis of Geophysical Properties, Unit Definitions, and Identifying the Incorrect Statement

Q.99: Which one of the following statements regarding geophysical exploration methods is NOT correct?

✓ **(a) Milligal is a unit of magnetic field.**

(b) Magnetic anomaly can be caused by pyrrhotite and magnetite.

(c) Gravity method of mineral exploration is based on density contrasts.

(d) Scintillometers can detect radioactivity in rocks.

Correct Option: (a) Milligal is a unit of magnetic field. (Statement is NOT correct)

1. Analysis of Statement (a): Units of Gravity vs. Magnetism

The statement that *"Milligal is a unit of magnetic field"* is **factually incorrect**:

- **The Gal & Milligal:** The **gal** (symbol Gal, named in honor of Galileo Galilei) is the CGS unit of **gravitational acceleration**:

$$1\ \text{Gal} = 1\ \text{cm/s}^2 = 10^{-2}\ \text{m/s}^2$$
$$1\ \text{milligal (mGal)} = 10^{-3}\ \text{Gal} = 10^{-5}\ \text{m/s}^2 \approx 10^{-6}\ \text{g}$$

Gravitational anomalies caused by mineral deposits, salt domes, and faults are typically measured in tenths to tens of **milligals (mGal)** or **gravity units (1 g.u. = 0.1 mGal = 1 μm/s^2)**.

- **Magnetic Units:** The magnetic field in geophysics (magnetic flux density *B*) is measured in **Tesla (T)** or **nanoTesla (nT)** in SI units, or traditionally in **gamma (γ)** and **Gauss (G)**:

$$1\ \text{nT} = 1\ \gamma = 10^{-9}\ \text{T} = 10^{-5}\ \text{Gauss}$$

Earth's total geomagnetic field ranges from ~25,000 nT near the magnetic equator to ~65,000 nT near the magnetic poles.

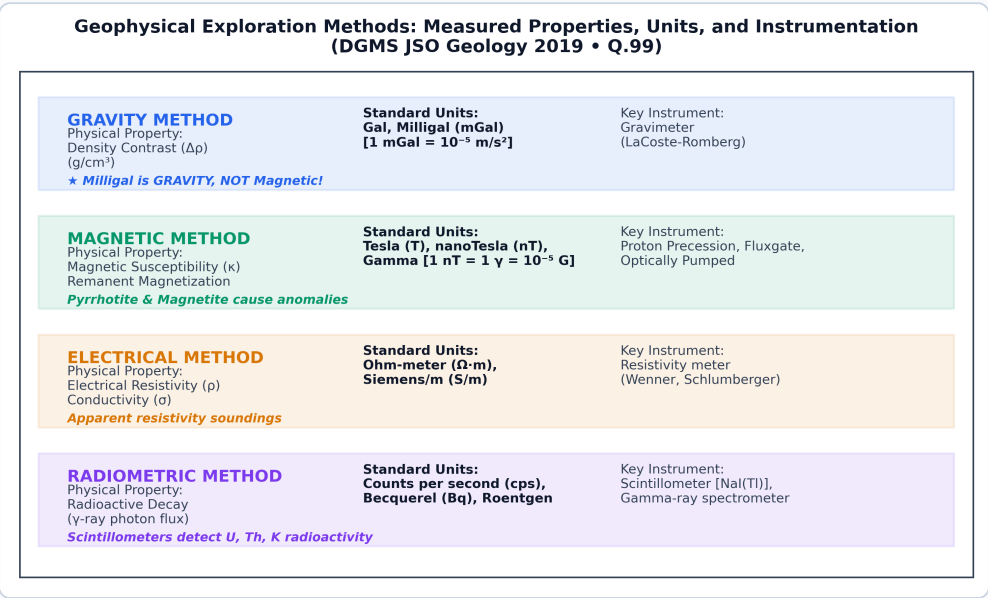


Figure 99.1: Comprehensive overview of geophysical exploration methods showing measured physical properties, standard units, and instrumentation. Notice that Milligal is strictly the unit of Gravity, not Magnetism. Based on Lowrie, Fundamentals of Geophysics and Telford et al., Applied Geophysics.

2. Verification of Correct Statements (b, c, d)

- **Statement (b) is CORRECT:** Both **magnetite (Fe_3O_4)** and monoclinic **pyrrhotite (Fe_7S_8)** are ferrimagnetic minerals possessing very high magnetic susceptibility and remanent magnetization. Pyrrhotite is a common constituent of nickel-copper sulfide deposits (e.g., Sudbury) and produces massive magnetic anomalies second only to magnetite.
- **Statement (c) is CORRECT:** The gravity method detects subsurface anomalies by measuring spatial variations in the gravitational field caused by lateral **density contrasts ($\Delta\rho = \rho_{target} - \rho_{host}$)**. High-density bodies like chromite, barite, and massive sulfides produce positive Bouguer gravity anomalies, while low-density salt domes produce negative anomalies.
- **Statement (d) is CORRECT:** A **scintillometer** is a radiometric instrument containing a thallium-activated sodium iodide crystal [NaI(Tl)] that emits light flashes (scintillations) when struck by gamma-ray photons emitted during the decay of radioactive isotopes (^{238}U, ^{232}Th, ^{40}K).

3. Systematic Option Evaluation

Option	Statement Evaluated	Factual Correctness	Status as Answer to Question
(a)	**Milligal is a unit of magnetic field.**	**INCORRECT.** Milligal (10^{-5} m/s^2) is the standard unit of gravitational acceleration, whereas magnetic field is measured in nanoTesla (nT) or gamma.	**Correct Choice (Identifies the incorrect statement)**
(b)	Magnetic anomaly can be caused by pyrrhotite and magnetite.	**CORRECT.** Magnetite and monoclinic pyrrhotite are strongly ferrimagnetic minerals that generate strong regional and local magnetic anomalies.	Incorrect Choice

Option	Statement Evaluated	Factual Correctness	Status as Answer to Question
(c)	Gravity method of mineral exploration is based on density contrasts.	**CORRECT.** The fundamental operating physical property of gravity prospecting is the lateral density contrast (Δρ) between the orebody and country rock.	Incorrect Choice
(d)	Scintillometers can detect radioactivity in rocks.	**CORRECT.** Scintillometers detect gamma-ray radiation from uranium, thorium, and potassium in radioactive prospecting.	Incorrect Choice

> **Key Geophysical Takeaway**
>
> In applied geophysics:
> • **Gravity:** Unit = **Milligal (mGal)** or $\mu m/s^2$ • Property = **Density (ρ)**
> • **Magnetics:** Unit = **nanoTesla (nT)** or gamma (γ) • Property = **Magnetic Susceptibility (κ)**

4. Standard Textbook References

- **William Lowrie**, *Fundamentals of Geophysics* (3rd Edition, Cambridge University Press, 2020), Chapter 2: "Gravity" and Chapter 5: "Earth's Magnetic Field", pp. 45–78, 280–315.
- **W. M. Telford, L. P. Geldart, & R. E. Sheriff**, *Applied Geophysics* (2nd Edition, Cambridge University Press), Chapter 2: "Gravity Methods" and Chapter 3: "Magnetic Methods", pp. 6–135.

- **Philip Kearey, Michael Brooks, & Ian Hill**, *An Introduction to Geophysical Exploration* (3rd Edition, Wiley-Blackwell), Chapter 6: "Gravity Surveying", pp. 102–138.